

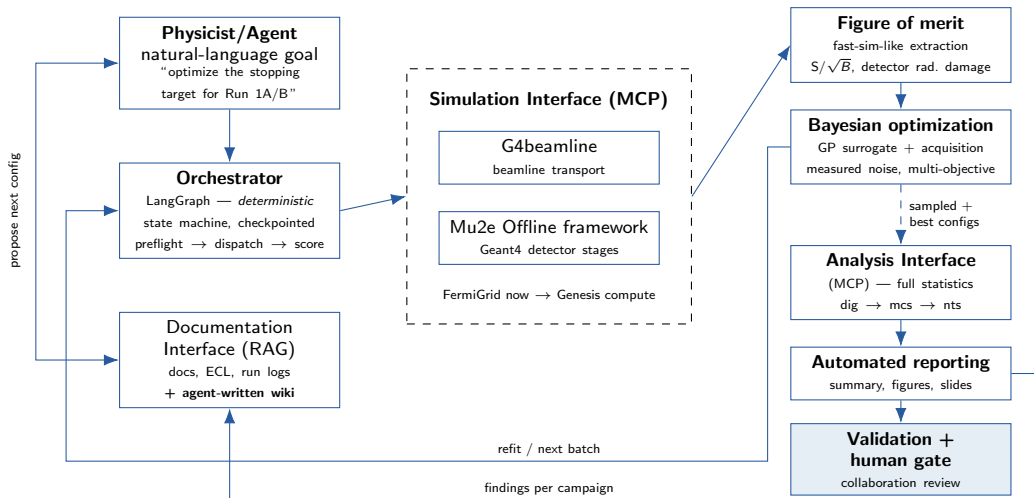
Mu2e Genesis: Agentic Experiment Optimization

Workflow (i) — activities and near-term plan

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Agentic Experiment Optimization — the loop

Prompt-to-configuration: simulate, analyze, score, decide — with a human gate.



Activities and owners (1/2)

Surrogate model optimization

(Yuri)

- ▶ Bayesian optimization: GP surrogate + multi-objective acquisition (sensitivity vs. detector radiation damage)
- ▶ Needed: multi-fidelity, transfer between scenarios
- ▶ Explore BO models for higher dimensionality: TuRBO, SAASBO, input warping

LangGraph orchestration

(Yuri / Ray / Rob)

- ▶ One graph run = one evaluation; checkpointed, resumable
- ▶ **Today: LangGraph + Bayesian optimization, prompted and orchestrated by Claude** — staying within Claude is a viable option
- ▶ Needed: expose through MCP, log traces
- ▶ Parameterize scenarios: beam intensity, solenoid field scale, arbitrary geometry configuration

Fast sim scoring, Run1A/B

(Michael)

- ▶ Today's only scoring path: $S/\sqrt{B}$ + radiation damage, no full reconstruction
- ▶ Needed: improve fast-sim performance; revisit Run 1B; agreement with full sim
- ▶ Explore other fast-sim options: beam optics, detector response

Activities and owners (2/2)

Full-sim validation (dig → mcs → nts)

(Sophie / James)

- ▶ Full chain, full statistics: best configs + every ~10th point, run async
- ▶ Needed: calibration vs. the fast score; evidence bar and sign-off

Infrastructure — MCP, LLM, computing

(Ray / Rob)

- ▶ MCP servers, agent skills, model endpoints, compute
- ▶ FermiGrid now → Genesis compute; token + CPU accounting

Documentation interface

(Cole / Simon)

- ▶ Search over docs, ECL, run and configuration logs
- ▶ Needed: context for the optimizer — which constraints are real

G4beamline

(Yuri)

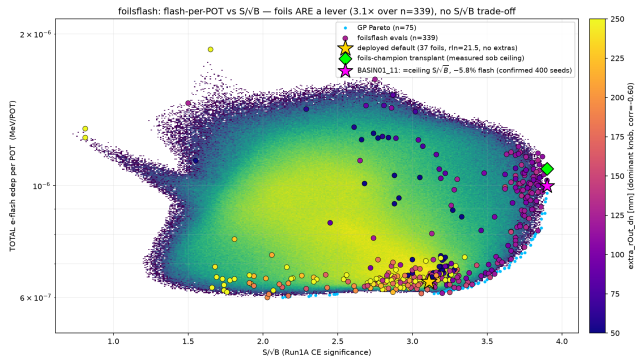
- ▶ MCP server prototype demonstrated — **a proposal commitment** (App. 5)
- ▶ Needed: integrate into the existing workflow

Automated reporting

(Rob / Yuri)

- ▶ Campaign drains → summary, figures, slides and write-up generated automatically
- ▶ Needed: regenerate on new rows; every number traceable to a leaderboard row
- ▶ **Write-back**: every campaign appends what it learned to the documentation interface — an agent-maintained wiki (Karpathy pattern); today hand-written per session

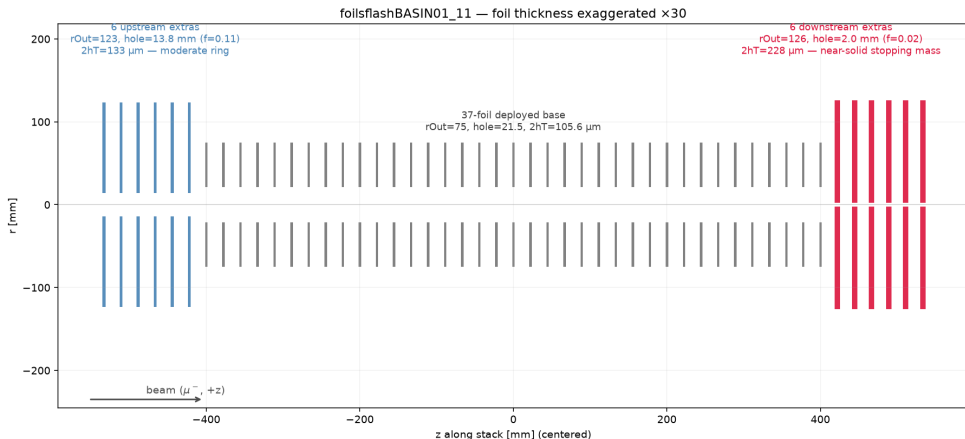
Status — the loop runs end-to-end



design	$S/\sqrt{B}$	rad. damage
deployed default	3.11	6.45×10^{-7}
ff11R00_07 (champion)	3.31	6.26×10^{-7}
ff14R01.00 (exploit)	3.84	8.14×10^{-7}
BASIN01.11 (confirmed)	3.90	1.00×10^{-6}

- ▶ **339** evaluations map a clean Pareto front
- ▶ Champion beats deployed on **both** axes: +6% $S/\sqrt{B}$ at -3% damage
- ▶ Best $S/\sqrt{B}$ **3.90 vs 3.11 deployed** = +25%, confirmed reproducible (400-seed)

What the optimizer built — the $\max\text{-}S/\sqrt{B}$ target



Two routes to high $S/\sqrt{B}$: a thin **off-beam ring**, or thin **in-beam** material that degrades the beam to stop more muons. Here: 6 upstream extras (moderate ring) + 6 downstream (near-solid stopping mass) around the 37-foil deployed base.

Phase I milestones — where Workflow (i) stands

9 months from **1 Aug 2026**; go/no-go at month 6 (Jan 2027). **We are in month 1.**

window	dates	milestone (Workflow (i) parts)	status
Months 1–2	Aug–Sep 26	Deploy interfaces: Simulation, Analysis, Documentation End-to-end agentic chain: sim → analysis → FoM	pending done
Months 3–4	Oct–Nov 26	Exercise optimization on campaigns; collect metrics	ahead
Months 5–6	Dec–Jan 27	Compile go/no-go results; all traces logged	traces = gap
Months 7–9	Feb–Apr 27	Harden interfaces; package artifacts for Genesis	—

- ▶ One month in: the **science milestone of months 3–4 is already met**; interfaces are the open item
- ▶ Proposal §4 anticipates this: *“initial workflow prototyping can use simplified interfaces while production versions are improved”*

Decision gate metrics — Workflow (i), proposal §6

(a) *“Closed-loop sim–analysis–optimization (agent-orchestrated search with Bayesian optimization for local sampling) demonstrated on ≥ 2 distinct external-constraint scenarios (e.g. half beam intensity, 10% reduced fields), each initiated by a natural-language prompt and completed within one week (vs. months for the manual baseline), with full traces logged”*

- **Started.** Prompt → new mode + 20 evaluations, saturation confirmed by a second campaign — a scenario answers **within a day** (bar: one week)
- Remaining: the two **external-constraint** scenarios — reduced / absent DS field, half beam intensity — and **formal trace logging**

(b) *“for a Run-1a scenario (no full cosmic ray veto coverage), starting from the collaboration’s existing manually-optimized baseline, agent-produced configuration achieves improved $R_{\mu e}$ sensitivity, with improvements documented for collaboration review”*

- **Demonstrated:** champion beats the deployed target on **both** axes ($+6\% S/\sqrt{B}$, -3% damage); **+25%** $S/\sqrt{B}$ at the confirmed ceiling
- To finish: **prove it with full sims** (dig → mcs → nts) on the best configs; scope by background regime; document for review

Backup

What “GP surrogate + multi-objective acquisition” means

GP surrogate — the map. Each real evaluation costs ~ 150 CPU-hours, so we cannot try many designs. The Gaussian Process looks at every design tried so far and draws a map of the whole space: for a design we have *not* tried it predicts the answer **and how unsure it is**. Near measured points the prediction is sharp; far away it is foggy. *The fog is the useful part.*

Acquisition — the rule for where to go next. Something has to choose which 10 designs to actually simulate next. The acquisition function trades off **exploit** (go where the map predicts good results) against **explore** (go where the fog is thick — we might be missing something).

Multi-objective — why there is no single “best”. We want two things at once: more signal, less detector damage. They fight each other, so there is no single winner — there is a **set** of best compromises, the **Pareto front** (the map on the status slide). Our acquisition function (qNEHVI) picks the designs expected to push that frontier *outward* the most, rather than chasing one number.